

SC3103: Embedded Programming

Tutorial 2: Real-Time Schedulability

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Notes: EDF is roughly based on $O(n \log T_i)$, where n is number of task, appearance period of task. can be derived from exact scheduling formula.

1. Consider the following periodic real-time task set:

$$\tau_1 = (3, 1, 3), \tau_2 = (5, 1, 5), \tau_3 = (7, 3, 7).$$

Which of the following scheduling algorithms, EDF or RM, can successfully schedule this task set. Justify your answer.

EDF(exact): $1/3 + 1/5 + 3/7 = 0.96 \leq 1$ thus its successful
RM (suff) = $0.96 \leq 3(2^{1/3} - 1) = 0.77$ (false), move on to exact, T2: $R_0 = C_2 = 1, R_1 = 1/3(1) + 1, R_2 = 2/3(1) + 1 = 2 \leq 5$. T3: $R_1 = 3/3(1) + 3/5(1) + 3 = 5, R_2 = 5/3(1) + 5/5(1) + 3 = 6, R_4 = 7/3(1) + 7/5(1) + 3 = 8 \geq D_3 = 7$ thus not meet deadline

2. Consider the following sporadic real-time task set.

$$\tau_1 = (15, 5, 15), \tau_2 = (17, 5, 17), \tau_3 = (19, 5, 19).$$

Is it schedulable under RM? Justify your answer

T3: $R_0 = 5, R_1 = 5/15(5) + 5/17(5) + 5 = 15, R_2 = 15/15(5) + 15/17(5) + 5 = 15 \leq 19$ thus can meet deadline. Do for T2 as well can meet deadline

3. Consider a task system comprising four tasks, $\tau_1 = (4, 1, 4), \tau_2 = (5, 3, 5), \tau_3 = (6, 2, 6)$ and $\tau_4 = (8, 5, 8)$, and scheduled on a **dual-core** platform.

- (a) Is this task system feasible? Justify your answer.
- (b) Identify all possible partitions of this set of tasks such that the resulting system is schedulable under both Partitioned Earliest Deadline First (EDF) and Partitioned Rate Monotonic (RM). Justify your answer.
- (c) Is this task system schedulable under Global Rate Monotonic (RM) scheduling algorithm? Justify your answer.
- (d) Construct the Global Earliest Deadline First (EDF) schedule for this task system up to time instant 15. Can you conclude from this schedule whether the task system is schedulable under Global EDF? Justify your answer.

EDF test is a feasible test for multi core, $1/4 + 3/5 + 2/6 + 5/8 \leq 2$ thus feasible.

b) any partitioning combination, must be able to see that T2 and T4 cant be together as each is more than 1/2 so when together it will be > 1 , and T1 T2 T3 together wont work, violate EDF test. THUS, when P1(T2 T1 and T4 T3) or P2(T2 T3 and T4 T1) will fulfill EDF as all of them ≤ 1 . for RM:

P1: T2: $R_0 = 3, R_1 = 3/4(1) + 3 = 4, R_2 = 4/4(1) + 3 = 4 \leq 5$, T4: $R_0 = 5, R_1 = 5/6(2) + 5 = 7, R_2 = 7/6(2) + 5 = 9 > 8$, wont hit deadline.

P2: T3: $R_0 = 2, R_1 = 2/5(3) + 2 = 5, R_2 = 5/5(3) + 2 = 5 \leq 6$, ok, T4: $R_0 = 5, R_1 = 5/4(1) + 5 = 7, R_2 = 7/4(1) + 5 = 7 \leq 8$, thus P2 will hit deadline for RM.